

Chapter 2 Homework Assignment

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Problem 1: Transformations and Standardization

Let X be a continuous random variable.

1. Prove $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

Sol.

$$\begin{aligned}\text{Var}(aX + b) &= E[(aX + b)^2] - E[(aX + b)]^2 \\ &= E[a^2 X^2 + 2abX + b^2] - [aE(X) + b]^2 \\ &= a^2 E[X^2] + 2abE[X] + b^2 - (a^2 E[X]^2 + 2abE[X] + b^2) \\ &= a^2 (E[X^2] - E[X]^2) \\ &= a^2 \text{Var}(X)\end{aligned}$$

2. Let $Y = \frac{X - \mu}{\sigma}$. Show that $E[Y] = 0$ and $\text{Var}(Y) = 1$.

Sol.

Let $\mu = E[X]$, $\sigma = \sqrt{\text{Var}(X)} > 0$

$$E[Y] = E\left[\frac{X - \mu}{\sigma}\right] = \frac{1}{\sigma}(E[X] - \mu) = \frac{1}{\sigma}(\mu - \mu) = 0$$

$$\text{Var}(Y) = \text{Var}\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \left(\frac{1}{\sigma}\right)^2 \text{Var}(X - \mu) = \frac{1}{\sigma^2} \text{Var}(X) = \frac{1}{\sigma^2} \sigma^2 = 1$$

3. If $X \sim \text{Exp}(\lambda)$ derive the CDF and verify the memoryless property.

Sol.

$$\text{CDF: } F_X(x) = P(X \leq x) = \int_0^x \lambda e^{-\lambda t} dt = [-e^{-\lambda t}]_0^x = 1 - e^{-\lambda x}$$

Thus

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0 \end{cases}$$

Let $s, t \geq 0$. Then

$$P(X > s + t \mid X > s) = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$$

Hence the exponential distribution is memoryless.

Problem 2: Normal Sampling Theory

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

1. Derive the distribution of $\bar{X}$.

Sol.

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} = \frac{X_1 + X_2 + \dots + X_n}{n}, \text{ then}$$

$$E[\bar{X}] = \frac{E[X_1 + X_2 + \dots + X_n]}{n} = \frac{n\mu}{n} = \mu$$

$$Var(\bar{X}) = \frac{Var[X_1 + X_2 + \dots + X_n]}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$$\text{Hence } \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

2. Show $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$.

Sol.

Let $S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$, and we already know the following:

(1) $\bar{X}$ and S^2 are independent by Basu's theorem.

(2) If $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$, then $Z^2 \sim \chi_1^2$.

(3) If $Y_i \stackrel{iid}{\sim} \chi_1^2$, then $\sum_{i=1}^n Y_i \sim \chi_n^2$, $i = 1, \dots, n$.

(4) If $Q \sim \chi_k^2$, then $M_Q(t) = (1 - 2t)^{-k/2}$ for $t < \frac{1}{2}$.

Compute

$$\begin{aligned} \sum_{i=1}^n (X_i - \mu)^2 &= \sum_{i=1}^n ((X_i - \bar{X}) + (\bar{X} - \mu))^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2. \end{aligned}$$

Then divide by σ^2 :

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2.$$

Let

$$U = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2, \quad V = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2, \quad W = \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)^2.$$

Then $U = V + W$, and we can know

$$U = \sum_{i=1}^n Z_i^2 \sim \chi_n^2, \quad V = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{(n-1)S^2}{\sigma^2},$$

also $\bar{X} \sim N(\mu, \sigma^2/n)$, hence

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \Rightarrow W \sim \chi_1^2.$$

By (1) and W is a function of $\bar{X}$ while $V = \frac{(n-1)S^2}{\sigma^2}$ is a function of S^2 , we have

$$V \perp\!\!\!\perp W.$$

From $U = V + W$ and independence, we have

$$M_U(t) = M_V(t) M_W(t).$$

By (4), we have

$$(1 - 2t)^{-n/2} = M_U(t) = M_V(t) (1 - 2t)^{-1/2} \Rightarrow M_V(t) = (1 - 2t)^{-(n-1)/2}$$

Since $M_V(t)$ is the mgf of χ_{n-1}^2 .

Finally, $V = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$.

3. Deduce $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$.

Sol.

Let $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$. From Q2, we know and let $U = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$.

Because Therefore

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \cdot \frac{\sigma}{S} = Z \cdot \frac{\sigma}{S}$$

$$\frac{\sigma}{S} = \left(\frac{S^2}{\sigma^2} \right)^{-1/2} = \left(\frac{U}{n-1} \right)^{-1/2} = \sqrt{\frac{n-1}{U}}.$$

所以

$$T = \frac{Z}{\sqrt{U/(n-1)}}.$$

依 t 分佈定義：若 $Z \sim N(0, 1)$ 、 $U \sim \chi_\nu^2$ 且獨立，則

$$\frac{Z}{\sqrt{U/\nu}} \sim t_\nu.$$

取 $\nu = n - 1$ ，得

Problem 3: Discrete Structure and Limits

1. Derive $E[X]$, $\text{Var}(X)$ for $X \sim \text{Bin}(n, p)$.
2. Prove the Poisson limit of Binomial.
3. Give an SPC context favoring Poisson modeling.

Problem 4: Mixtures and Overdispersion

1. Show Poisson–Gamma $\Rightarrow$ Negative Binomial.
2. Show Beta–Binomial via marginalization.
3. Explain why mixtures increase variance.